

Chapter 1

Computer Abstractions and Technology

1.1 The Computer Revolution

- Progress in computer technology
 - Underpinned by Moore's Law
- Makes novel applications feasible
 - Computers in automobiles
 - Cell phones
 - Human genome project
 - World Wide Web
 - Search Engines
- Computers are pervasive

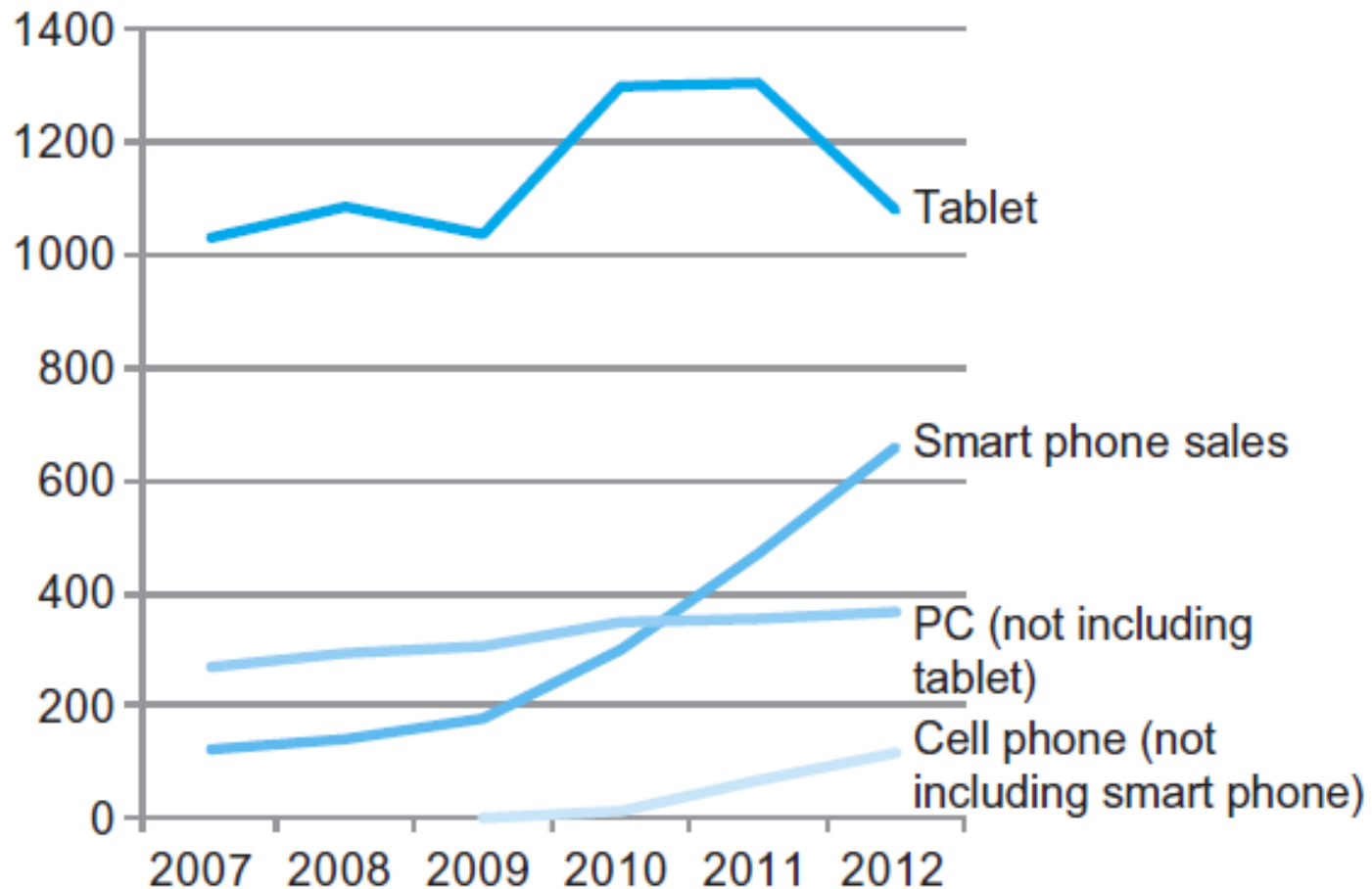
1.1 Classes of Computers

- Personal computers
 - General purpose, variety of software
 - Subject to cost/performance tradeoff
- Server computers
 - Network based
 - High capacity, performance, reliability
 - Range from small servers to building sized

1.1 Classes of Computers

- Supercomputers
 - High-end scientific and engineering calculations
 - Highest capability but represent a small fraction of the overall computer market
- Embedded computers
 - Hidden as components of systems
 - Stringent power/performance/cost constraints

1.1 The PostPC Era



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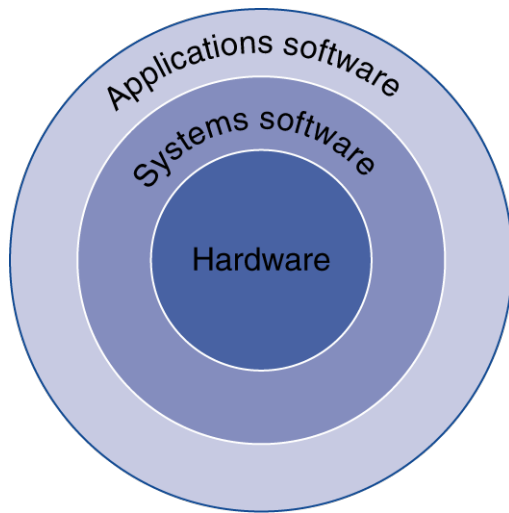
- Personal Mobile Device (PMD)
 - Battery operated
 - Connects to the Internet
 - Hundreds of dollars
 - Smart phones, tablets, electronic glasses
- Cloud computing
 - Warehouse Scale Computers (WSC)
 - Software as a Service (SaaS)
 - Portion of software run on a PMD and a portion run in the Cloud
 - Amazon and Google

1.2 Eight Great Ideas

- Design for **Moore's Law**
- Use **abstraction** to simplify design
- Make the **common case fast**
- Performance via **parallelism**
- Performance via **pipelining**
- Performance via **prediction**
- **Hierarchy** of memories
- **Dependability** via redundancy



1.3 Below Your Program



- Application software
 - Written in high-level language
- System software
 - Compiler: translates HLL code to machine code
 - Operating System: service code
 - Handling input/output
 - Managing memory and storage
 - Scheduling tasks & sharing resources
- Hardware
 - Processor, memory, I/O controllers

1.3 Levels of Program Code

- High-level language
 - Level of abstraction closer to problem domain
 - Provides for productivity and portability
- Assembly language
 - Textual representation of instructions
- Hardware representation
 - Binary digits (bits)
 - Encoded instructions and data

High-level
language
program
(in C)

```
swap(int v[], int k)
{int temp;
  temp = v[k];
  v[k] = v[k+1];
  v[k+1] = temp;
}
```

Compiler

Assembly
language
program
(for RISC-V)

```
swap:
    slli x6, x11, 3
    add  x6, x10, x6
    ld   x5, 0(x6)
    ld   x7, 8(x6)
    sd   x7, 0(x6)
    sd   x5, 8(x6)
    jalr x0, 0(x1)
```

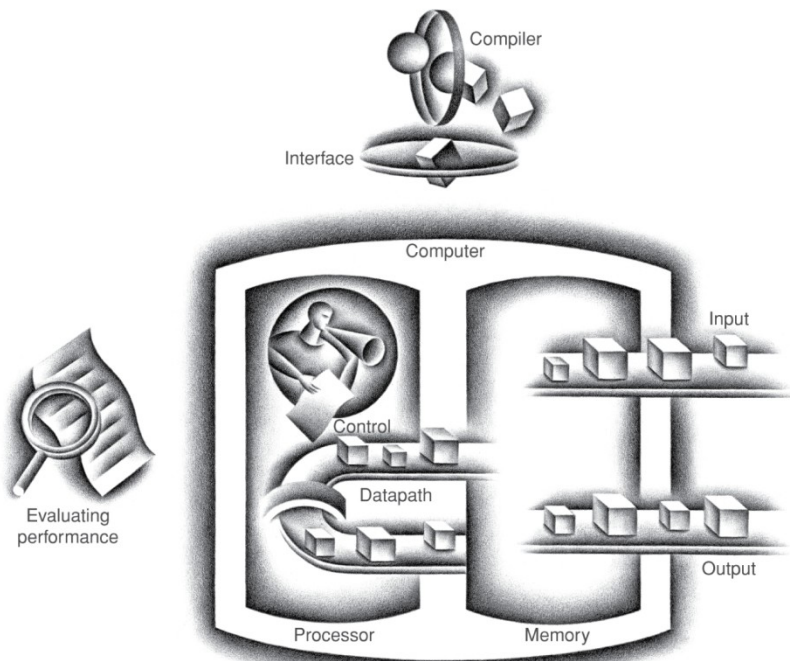
Assembler

Binary machine
language
program
(for RISC-V)

```
00000000001101011001001100010011
00000000011001010000001100110011
00000000000000110011001010000011
00000000100000110011001110000011
00000000011100110011000000100011
00000000010100110011010000100011
00000000000000001000000011001111
```

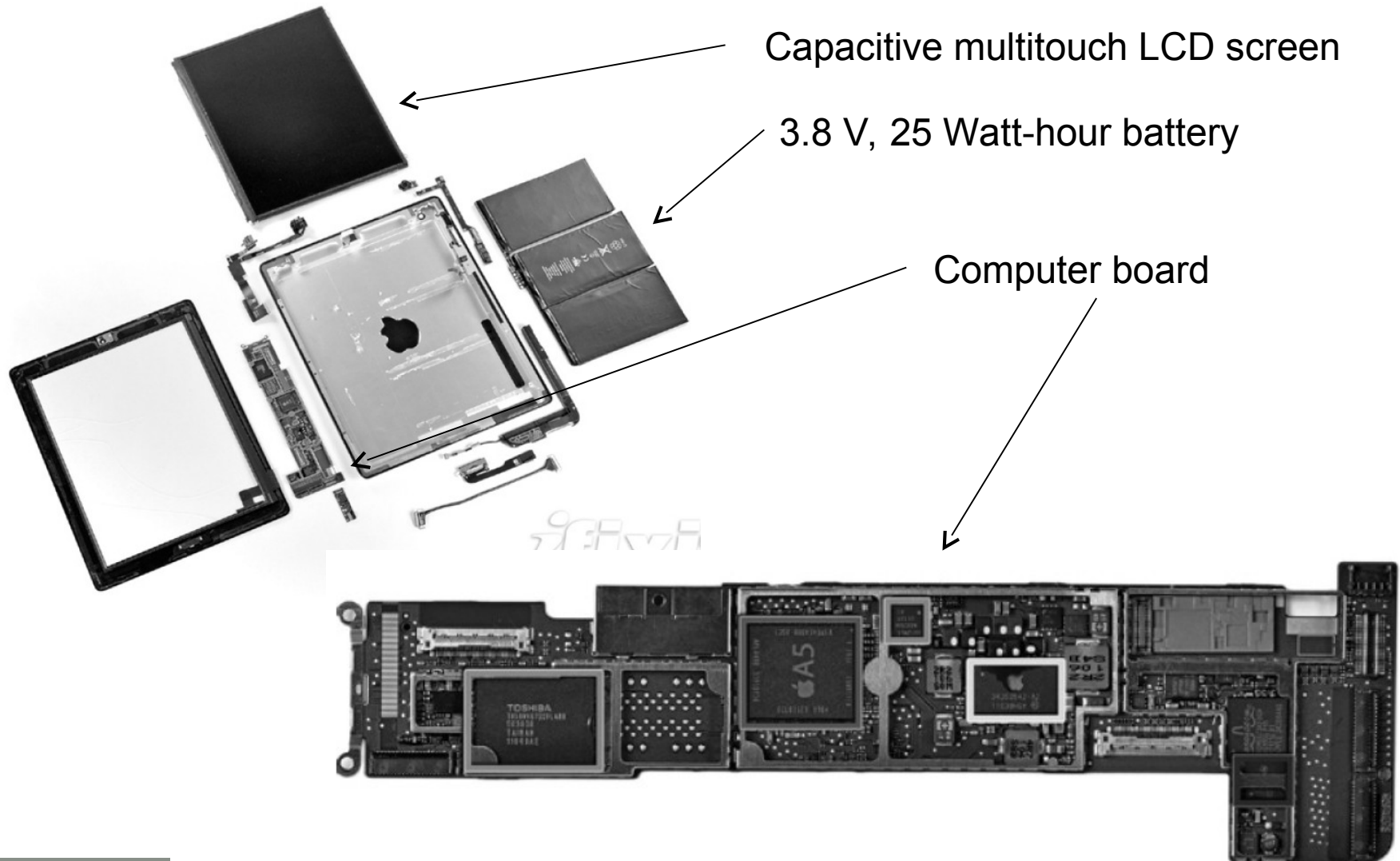
1.4 Components

The BIG Picture



- Same components for all kinds of computer
 - Desktop, server, embedded
- Input/output includes
 - User-interface devices
 - Display, keyboard, mouse
 - Storage devices
 - Hard disk, CD/DVD, flash
 - Network adapters
 - For communicating with other computers

1.4 Opening the Box



1.4 Inside the Processor

- Datapath: performs operations on data
- Control: sequences datapath, memory, ...
- Cache memory
 - Small fast SRAM memory for immediate access to data

1.4 Inside the Processor

- Apple A5



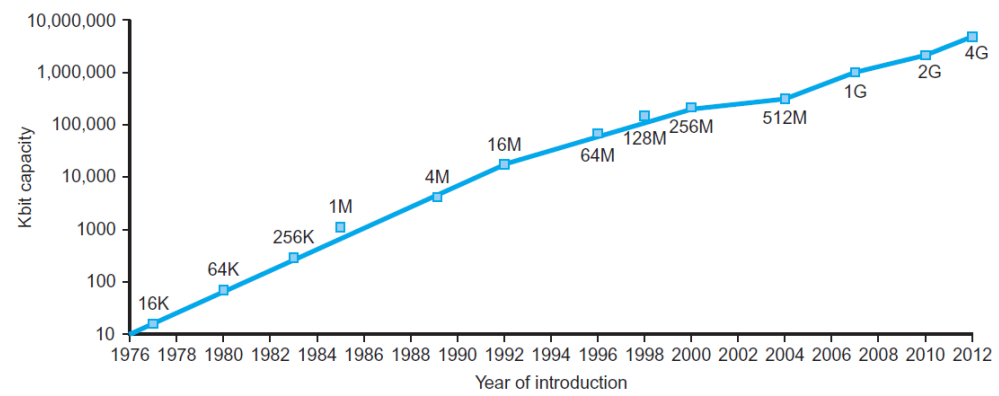
1.4 Abstractions

The BIG Picture

- Abstraction helps us deal with complexity
 - Hide lower-level detail
- Instruction set architecture (ISA)
 - The hardware/software interface
- Application binary interface
 - The ISA plus system software interface
- Implementation
 - The details underlying and interface

1.5 Technology Trends

- Electronics technology continues to evolve
 - Increased capacity and performance
 - Reduced cost



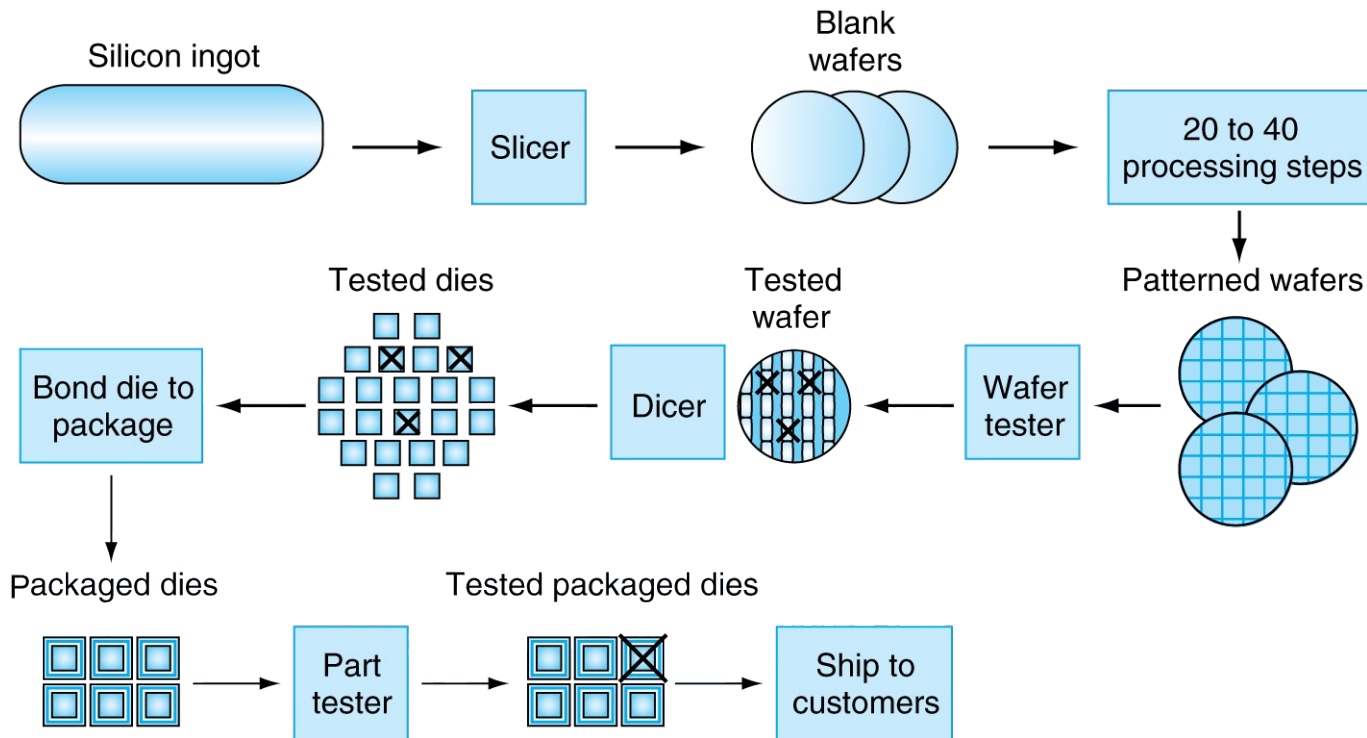
DRAM capacity

Year	Technology	Relative performance/cost
1951	Vacuum tube	1
1965	Transistor	35
1975	Integrated circuit (IC)	900
1995	Very large scale IC (VLSI)	2,400,000
2013	Ultra large scale IC	250,000,000,000

1.5 Semiconductor Tech.

- Silicon: semiconductor
- Add materials to transform properties:
 - Conductors
 - Insulators
 - Switch

1.5 Manufacturing ICs



- **Yield**: proportion of working dies per wafer